Math 152 Lecture Notes: Week 1

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1 Introduction

1.1 Course Format and Logistics

1. My name is Miles Kretschmer, I am a PhD student here.
2. My email is `mjkretschmer@uchicago.edu`.
3. My office is Eckhart 7, in the basement. Office hours will be decided this week. Feel free to email if you want to set up a meeting in the meantime.
4. Information about course policies can be found on the syllabus. Briefly, there will be weekly homework assignments, with a computational part on the Edfinity platform as well as a set of longer form problems. The first assignment will be released on Wednesday. Everything can be accessed on canvas. There will be two midterms and a final.

1.2 Motivation

This is a second course in calculus, which focuses primarily on integration and transcendental functions. Integration will allow us to calculate the area of plane regions, the volume of solids, and centers of mass of objects. It also lets us infer the total accumulated change in a quantity given its rate of change, for instance the displacement of an object given its velocity. To achieve these ends, we will learn many methods of computing integrals.

Transcendental functions are those which cannot be defined in terms of algebraic operations such as addition, multiplication, and extraction of roots of polynomials. We will see that several such functions: the logarithm, exponential, and inverse trigonometric functions, arise naturally from integrals involving algebraic functions. This way of introducing these functions explains some of their remarkable properties in terms of properties of integrals.

The endpoint of the course will be to apply integration and transcendental functions to find solutions to differential equations, equations that relate a quantity to its rates of change. A wide variety of quantitative laws considered in natural science can be expressed in this way, including Newton's laws of motion, the laws governing the behavior of electric and magnetic fields, the flow of heat, population growth, and rates of chemical reactions. Finding explicit solutions to such equations is one way of drawing predictions out of these laws.

1.3 Plan for This Week

We will begin by going back to the beginning of calculus, the limit, and looking at it with new eyes. We will learn how to formally define the limit of a function at a point, and what it means for a function to be continuous. After this, we will review the main points of differential calculus before we begin with integration.

2 Limits and Continuity

A brief historical note: calculus itself originates in the 17th century, but the way of treating limits that we will learn was not made explicit until the 19th century. It was (and still is) seen as a way of clarifying the foundations of calculus, which were originally viewed with suspicion. It took a long time for these ideas to solidify, so don't worry if you find them challenging at first.

The operations of calculus, differentiation and integration, can both be defined in terms of limits. You have probably seen the derivative defined follows.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

In general, the expression

$$\lim_{x \to a} f(x) = L$$

informally means that as x approaches a, $f(x)$ approaches L. This can make sense even if f is not defined at a, but on the other hand, the limit of f at a may not exist in general. [Draw pictures exhibiting various kinds of existence and nonexistence of limits.]

How to we make this into a precise mathematical definition?

2.1 The Definition of the Limit

We will define the limit by making the qualitative statements "x approaches a" and "$f(x)$ approaches L" into quantitative statements, by considering how the distance of $f(x)$ to L depends on the distance of x to a. The distance between two real numbers x, y is $|x - y|$, so the inequality $|f(x) - L| < \varepsilon$ means that $f(x)$ is within a distance of ε from L, where $\varepsilon > 0$ is some allowed error. Similarly, $|x - a| < \delta$ means that x is within a distance δ of a.

We interpret "as x approaches a, $f(x)$ approaches L" to mean "given any allowed error, we can ensure that $f(x)$ is within that distance of L by taking x sufficiently close to a."

Definition. Suppose that f is a function defined on an interval around a, but maybe not defined at a itself. We say that

$$\lim_{x \to a} f(x) = L$$

if for any $\varepsilon > 0$, there is a $\delta > 0$ such that whenever $0 < |x - a| < \delta$, $|f(x) - L| < \varepsilon$.

[Draw picture illustrating this.]

Notice that in this definition, δ is allowed to depend on ε: as we make ε smaller, we will in general need to make δ smaller as well. Notice also that if some δ works, then so does any smaller δ.

Example 1. Let's verify that $\lim_{x \to 1}(2x + 1) = 3$ according to our definition. Substituting in the function and numbers here, we need to check that the following is true: for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - 1| < \delta$, then $|2x + 1 - 3| < \varepsilon$. One way to think of this kind of argument is to imagine that someone (our opponent) gives us a positive number $\varepsilon > 0$, we cannot control what it is. We then have to choose a number δ satisfying the limit definition. If we can choose a δ for any ε they give us, we win. To see how to choose δ, we work backwards from the inequality we want:

$$|2x + 1 - 3| < \varepsilon$$

$$\rightsquigarrow |2x - 2| < \varepsilon$$

$$\rightsquigarrow 2|x - 1| < \varepsilon$$

$$\rightsquigarrow |x - 1| < \varepsilon/2$$

This suggests that given ε, we let $\delta = \varepsilon/2$. Then running the above computations backward, whenever $|x-1| < \delta = \varepsilon/2$, we have

$$2|x-1| < \varepsilon \rightsquigarrow |2x-2| < \varepsilon \rightsquigarrow |2x+1-3| < \varepsilon$$

as required. We win!

The above method of choosing δ in terms of ε works for any linear function.

Proposition. *If $f(x) = ax + b$, then for any c we have $\lim_{x\to c}(ax+b) = ac+b$.*

Proof sketch. We need to check that for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x-c| < \delta$, then $|ax+b-(ac+b)| < \varepsilon$. I leave it to you to check that for $a \neq 0$, $\delta = \varepsilon/|a|$ works, and for $a = 0$, any δ whatsoever works. □

Example 2. We can explain with this definition why $\lim_{x\to 0}\sin(x^{-1})$ does not exist. [Draw picture]. No matter how small δ is, the function $\sin(x^{-1})$ goes all the way up to 1 and all the way down to -1 within the interval $(-\delta, \delta)$, so it can never be confined to within ε of any limit for $\varepsilon < 1$.

Notice that in our definition of the limit, we consider values $f(x)$ where x is close to a, but not $f(a)$ itself (which may not even be defined). This means that if $f(x) = g(x)$ for all x in an interval around a, except maybe a itself, then $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$ (one exists if and only if the other exists.) [Draw picture with removable discontinuity]

Example 3. The limit defining the derivative of x^2 is

$$\lim_{h\to 0}\frac{(a+h)^2 - a^2}{h} = \lim_{h\to 0}\frac{2ah+h^2}{h}$$

The expression

$$\frac{2ah+h^2}{h}$$

is undefined at $h = 0$, but is equal to $2a + h$ for all $h \neq 0$. Therefore, the limit is

$$\lim_{h\to 0}(2a+h) = 2a$$

Example 4. Consider

$$\lim_{x\to 1}\frac{x^2-1}{x-1}$$

We can factor the numerator as $(x+1)(x-1)$, and so for $x \neq 1$ this is equal to $x+1$. Thus the limit is 2.

2.2 Left and Right Limits

It will sometimes be convenient to break up our consideration of a limit into two pieces. First we consider $f(x)$ for $x < a$, and then $f(x)$ for $x > a$. In terms of these, we define the left and write limit, as follows.

Definition (Left limit)**.** We say that

$$\lim_{x\to a^-} f(x) = L$$

if for any $\varepsilon > 0$, there is a $\delta > 0$ such that for $a - \delta < x < a$, $|f(x) - L| < \varepsilon$.

In other words, $f(x)$ approaches L as x approaches a from the left. [Draw picture]

Definition (Right limit)**.** We say that

$$\lim_{x\to a^+} f(x) = L$$

if for any $\varepsilon > 0$, there is a $\delta > 0$ such that for $a < x < a + \delta$, $|f(x) - L| < \varepsilon$.

In other words, $f(x)$ approaches L as x approaches a from the right.

Proposition. *$\lim_{x\to a} f(x) = L$ if and only if $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$.*

[Draw pictures]

2.3 Continuity

You have probably seen the following definition of continuity.

Definition. A function f is continuous at a point a if $\lim_{x\to a} f(x)$ exists and is equal to $f(a)$. We say that f is continuous if it is continuous at every point.

This will be our definition as well, but we can reinterpret it in terms of our definition of limit.

Definition (Equivalent form)**.** A function f is continuous at a point a if for every $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$.

Informally, if we want the output to remain close to $f(a)$, we can accomplish this by taking the input sufficiently close to a. Small changes in input produce small changes in output. [Draw picture] Our argument for the limit of a linear functions implies the following.

Proposition. *Linear functions $f(x) = ax + b$ are continuous (everywhere).*

We will see that many more familiar functions are continuous.

Example 5. Consider the function f, with the following piecewise definition:

$$f(x) = \begin{cases} -1 & x \le 0 \\ 1 & x > 0 \end{cases}$$

[draw graph] I claim that this is not continuous at 0. Take $\varepsilon = 1$. For any $\delta > 0$, the interval $(-\delta, \delta)$ around 0 contains a positive number x, for which $f(x) = 1$ and so $|f(x) - f(0)| = 2 > \varepsilon$.

Here is a somewhat trickier example.

Example 6. Consider the function $f(x) = x^2$. We will show that this is continuous at every real number a. We need to show that for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|x^2 - a^2| < \varepsilon$. The idea is that we can factor $x^2 - a^2$ as $|x - a| \cdot |x + a|$. We can then control the size of both of these terms. As long as $|x - a| < 1$, $|x + a| < 2|a| + 1$. Then, $|x^2 - a^2| < |x - a| \cdot (2|a| + 1)$. We will then choose

$$\delta = \min\left(1, \frac{\varepsilon}{2|a| + 1}\right)$$

Then if $|x - a| < \delta$,

$$|x^2 - a^2| = |x + a| \cdot |x - a| < (2a + 1)|x - a| < \varepsilon$$

as required.

Example 7. Here is an even trickier example. Let $f(x) = \sqrt{x}$. We will first show that this is continuous at any $a > 0$. That is, for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$. This time we use the factorization

$$|\sqrt{x} - \sqrt{a}| \cdot |\sqrt{x} + \sqrt{a}| = |x - a|$$

and so

$$|\sqrt{x} - \sqrt{a}| = \frac{|x - a|}{|\sqrt{x} + \sqrt{a}|} < \frac{|x - a|}{\sqrt{a}}$$

provided that $x \ge 0$ so $\sqrt{x}$ is well defined. Thus if we want to ensure that $|\sqrt{x} - \sqrt{a}| < \varepsilon$, we need $|x - a| < |a|$, so $x \ge 0$, and $|x - a| < \sqrt{a} \cdot \varepsilon$, so that

$$\frac{|x - a|}{\sqrt{a}} < \varepsilon$$

Taking $\delta = \min(a, \sqrt{a}\varepsilon)$ works.

The function $f(x) = \sqrt{x}$ is also continuous at 0, in the sense that $\lim_{x\to 0^+} f(x) = 0 = f(0)$ (the function is not defined for $x < 0$, so this is all we can do.) Indeed, given any $\varepsilon > 0$, if $x > 0$ and $x < \varepsilon^2$, then $|\sqrt{x}| < \varepsilon$.

For the foundations of integration, we will need the following fact about continuous functions.

Theorem 1. *If f is a continuous function on a closed interval $[a, b]$, then f attains an absolute minimum and an absolute maximum on this interval. That is, there are $x_{\min}$ and $x_{\max}$ in $[a, b]$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ for all $x \in [a, b]$.*

[Draw picture.]

3 Methods of Evaluating Limits

We will not need to go back to ε's and δ's whenever we need to verify the limit of a new function; there are lots of ways to infer new limits from already established ones, that you have likely seen in your previous experience with calculus. Here I will review some of these limit laws and comment on how they can be grounded in the ε, δ definition. These limit laws in turn imply that certain combinations of continuous functions are continuous.

Proposition. *Suppose that $\lim_{x\to a} f(x) = K$ and $\lim_{x\to a} g(x) = L$.*

1. $\lim_{x\to a}(f(x) + g(x)) = K + L$
2. $\lim_{x\to a}(c \cdot f(x)) = c \cdot K$ *where c is a constant.*
3. $\lim_{x\to a}(f(x) \cdot g(x)) = K \cdot L$
4. *If $L \neq 0$,* $\lim_{x\to a}(f(x)/g(x)) = K/L$

Sketch of 1. Given ε, we can choose δ small enough that if $0 < |x-a| < \delta$, $|f(x)-K| < \varepsilon/2$ and $|g(x)-L| < \varepsilon/2$. Then

$$|f(x) + g(x) - (K + L)| = |(f(x) - K) + (g(x) - L)| \leq |f(x) - K| + |g(x) - L| < \varepsilon$$

Using the *triangle inequality* $|y + z| \leq |y| + |z|$. □

The proposition translates into the following properties of continuity.

Proposition. *Suppose f and g are functions:*

1. *$f + g$ is continuous wherever f and g are both continuous*
2. *$c \cdot f$ is continuous wherever f is continuous*
3. *$f \cdot g$ is continuous wherever f and g are both continuous*
4. *f/g is continuous wherever f and g are both continuous and $g \neq 0$.*

Example 8. Any polynomial function can be built as a sum of products of constants and powers of x. Using 1,2,3 and the fact that $f(x) = x$ is continuous, any polynomial is continuous (everywhere). A *rational function* $h(x) = f(x)/g(x)$, where f and g are polynomials, is continuous wherever $g \neq 0$, using 4.

Proposition. *Suppose $\lim_{x\to a} f(x) = b$ and $\lim_{y\to b} g(y) = L$. Then $\lim_{x\to a} g(f(y)) = L$.*

We write $g \circ f$ for the function sending x to $g(f(x))$.

Proposition. *If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a.*

Question. Where is the function

$$f(x) = \frac{x^3 + 2}{x\sqrt{1+x}}$$

defined? Where is it continuous?

Answer. For $f(x)$ to be define we must have $1 + x \geq 0$ and $x\sqrt{1+x} \neq 0$. This happens exactly for x in $(-1, 0) \cup (0, \infty)$. The function f can be built up from polynomials and square roots by composition, products, and quotients, so it is continuous where it is defined. □

Theorem 2 (Squeeze Theorem)**.** *If f, g, h are functions on an interval $(a - d, a + d)$ around a (maybe not defined at a), $f(x) \leq g(x) \leq h(x)$ for x in this interval, and*

$$\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$$

then $\lim_{x \to a} g(x)$ exists and is equal to L.

Proof Sketch. Given $\varepsilon > 0$, take a δ small enough that it works for both $f(x)$ and $h(x)$. That is, for x within a distance δ of a, both $f(x)$ and $h(x)$ are within ε of L. Because $g(x)$ is between $f(x)$ and $h(x)$, it must also be within ε of L. □

Example 9. The function $g(x) = x^2 \sin(x^{-1})$ is squeezed between $-x^2$ and x^2, and so $\lim_{x \to 0} g(x) = 0$. [Draw picture.]

4 Differential Calculus

Recall that the derivative of a function is defined as a limit of difference-quotients:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

This represents the instantaneous rate of change of $f(x)$ with respect to x. If $f'(x)$ is positive in some region, f is increasing, and if $f'(x)$ is negative, f is decreasing.

Given a function f, if this limit exists for all a in some set, we get another function f' defined on this set, also denoted $\frac{\mathrm{d}}{\mathrm{d}x} f(x)$. In this section, we will briefly review differentiation, the process of going form f to f', and get back into the practice of computing derivatives.

First, of all, differentiation satisfies the following properties:

$$(f + g)' = f' + g'$$

$$(c \cdot f)' = cf'$$

for c a constant. Together with the power rule,

$$\frac{\mathrm{d}}{\mathrm{d}x} x^n = nx^{n-1}$$

we can use these rules to compute the derivative of any polynomial.

Example 10. What is the derivative of $f(x) = 3x^5 - 2x + x^3$? Pulling out the constants and applying the power rule to each term, we get $f'(x) = 15x^4 - 2 + 3x^2$.

The power rule applies to fractional powers $x^{p/q}$ as well:

$$\frac{\mathrm{d}}{\mathrm{d}x}x^{p/q} = \frac{p}{q}x^{p/q-1}$$

In particular the derivative of $\sqrt{x} = x^{1/2}$ is $1/2\sqrt{x} = (1/2)x^{-1/2}$.

We also have the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)\cdot g(x)) = f'(x)\cdot g(x) + f(x)\cdot g'(x)$$

The quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)/g(x)) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

and the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(g(f(x)) = g'(f(x))\cdot f'(x)$$

Using these, we can compute derivatives of functions built out of other functions.

Example 11. We will compute the derivative of

$$\frac{x}{\sqrt{1+2x}}$$

First we write this as $g(x)/h(x)$, where $g(x) = x$ and $h(x) = \sqrt{1+2x}$. Using the quotient rule, we have

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2} = \frac{\sqrt{1+2x} + xh'(x)}{1+2x}$$

To compute $h'(x)$, we use the chain rule:

$$h'(x) = \frac{2}{2\sqrt{1+2x}} = \frac{1}{\sqrt{1+2x}}$$

Thus we obtain

$$f'(x) = \frac{\sqrt{1+2x} - \frac{x}{\sqrt{1+2x}}}{1+2x}$$

We have the following rule for differentiating trigonometric functions.

$$\sin'(x) = \cos(x)$$

$$\cos'(x) = -\sin(x)$$

Remark. These can be justified from the limit definition of the derivative, using the geometric definitions of sine and cosine and the squeeze theorem. This is outside the scope of this review, but check out Sections 2.5 and 3.6 of the text if you are curious.

Question. What is $\tan'(x)$?

Answer. We have $\tan(x) = \sin(x)/\cos(x)$. Using the quotient rule, we obtain

$$\tan'(x) = \frac{\cos(x)^2 - (-\sin(x)^2)}{\cos(x)^2} = \frac{1}{\cos(x)^2}$$

□

Remark. You have likely also seen rules for differentiating the exponential and logarithm functions. I will leave this out for now, as later in the course we will construct these functions from scratch. The rules to differentiate them will fall out of this construction.

Another thing left out at this point is the derivative of the inverse of a function. This will be also be discussed later in the course.

4.1 Questions for Group Work

Question. Let $f(x) = -x^3 + 3x^2 + 9x - 1$. On what intervals is f increasing/decreasing?

Question. Compute $f'(x)$, where

$$f(x) = \frac{2 + (1-x)^{-2}}{1 + 2x^2\sqrt{1+x^{-4}}}$$

Question. Let $f(x) = x\sin(1/x)$. Describe what happens to $f'(x)$ as x approaches 0. What about $g'(x)$ where $g(x) = x^2\sin(1/x)$?

5 Definite Integrals

Now we are ready to begin with the main topic of the course, the integral. You may have some familiarity with this already, but we will start from the foundations and build up to the point of computing integrals.

We start by considering an instance of integration which has a straightforward interpretation, the definite integral of a continuous function f which is nonnegative, $f(x) \geq 0$ for $x \in [a, b]$. This is denoted

$$\int_a^b f(x)\mathrm{d}x$$

and represents the area A between the x-axis $y = 0$ and the graph of f. [Draw picture.] As with the limit before, we will translate this geometric description into a definition in terms of the values of f. Our approach to defining the integral will be to put upper and lower bounds on this area, in such a way that we can make these bounds as close together as we want. We then define the integral to be the unique number below every upper bound and above every lower bound.

In order to approximate the integral, we divide the interval $[a, b]$ into smaller subintervals, and estimate the area above each subinterval using rectangles. Specicically, we choose points $x_0 < x_1 < \cdots < x_n$ in $[a, b]$, with $a = x_0$ and $x_n = b$, and consider the intervals $[x_{i-1}, x_i]$ for $i = 1, 2, \ldots, n$. Such a division of the interval into subintervals is called a partition

$$\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_{n-1} < x\}$$

On each interval in the partition $[x_{i-1}, x_i]$, f attains a minimum $f(x_{\min}) = \ell_i$ and a maximum $f(x_{\max}) = u_i$. The area of the region above each $[x_{i-1}, x_i]$ is bounded below by $\ell_i(x_i - x_{i-1})$ and bounded above by $u_i(x_i - x_{i-1})$. [Draw picture]. Letting $\Delta x_i = x_i - x_{i-1}$, we have the following estimates on the area A.

$$L(f, \mathcal{P}) = \sum_{i=1}^{n} \ell_i \Delta x_i \leq A \leq \sum_{i=1}^{n} u_i \Delta x_i = U(f, \mathcal{P})$$

The expressions labeled $L(f, \mathcal{P})$ and $U(f, \mathcal{P})$ are called lower and upper sums, respectively.

To define the integral, we consider all possible partitions of the interval. [Draw picture with lots of lower and upper bounds converging on a value.]

Proposition. *Provided that f is continuous, than one can choose a partition that makes the upper and lower bounds as close together as we want. That is, for any $\varepsilon > 0$, there is a partition $\mathcal{P}$ such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.*

Thus considering all the upper and lower bounds, we converge on a unique value for the area.

Definition. The integral of f over $[a, b]$,

$$\int_a^b f(x)\mathrm{d}x$$

is the unique number above every lower bound $L(f, \mathcal{P})$ and below every upper bound $U(f, \mathcal{P})$.

Notice that this definition makes sense for any continuous function, even those which take negative values. The geometric interpretation changes however. For a general continuous f,

$$\int_a^b f(x)\mathrm{d}x$$

represents the area above the x-axis minus the area below the x-axis. [Draw picture with colors.]

Example 12. Let $f(x) = c$ be constant. Then for any partition $\mathcal{P}$ of $[a, b]$, we have $\ell_i = u_i = c$ for all i, and so

$$U(f, \mathcal{P}) = L(f, \mathcal{P}) = \sum_i c \cdot \Delta x_i = c \sum_i \Delta x_i = c(b - a)$$

Therefore,

$$\int_a^b c\mathrm{d}x = c(b - a)$$

as expected. [Draw picture]

Remark. It is not really tractable to apply this definition to compute integrals of all but the most simple functions. We will later learn a much more effective way to compute integrals. It is, however, important that the integral is defined like this. This definition comes from the interpretation of the integral as an area in the case we are considering, and underlies other interpretations of the integral in geometry, physics, and beyond.

Next time we will consider another interpretation of this definition of the integral: recovering displacement from instantaneous velocity. This will be a prototype of the fundamental theorem of calculus, which relates differentiation and integration and provides the most effective way to compute integrals.

Remark. One does not need to consider the absolute maximum and absolute minimum to get a good approximation of the integral. For any choice of sample point $t_i \in [x_{i-1}, x_i]$, we have

$$L(f, \mathcal{P}) \leq \sum_{i=1}^{n} f(t_i)\Delta x_i \leq U(f, \mathcal{P})$$

so if $L(f, \mathcal{P})$ and $U(f, \mathcal{P})$ are close together (and so close to the true value of the integral) the sum in the middle will also be close to the true value. In fact, one can show that for any $\varepsilon > 0$, for any sufficiently fine partition and any choice of t_i, the sum above is within ε of the true value of the integral. This provides another way of defining the integral.